

Boundary–Horizon Law (Structural Edition)

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Abstract

The Boundary–Horizon Law formalises conservation and transfer of structural information across boundaries and horizons. It defines boundary flux, horizon flux, conservation surfaces, collapse-transfer modes, recurrence-transfer modes, motif-boundary coupling, gradient-horizon deformation, and regime-dependent boundary–horizon behaviour. This edition presents the basis-independent formulation suitable for academic publication and cross-domain application.

1 Introduction

The Boundary–Horizon Law describes how structural information behaves when crossing boundaries and horizons. It governs:

- Boundary Flux
- Horizon Flux
- Conservation Surfaces
- Collapse-Transfer Modes
- Recurrence-Transfer Modes
- Motif-Boundary Coupling
- Gradient-Horizon Deformation
- Regime-Dependent Boundary–Horizon Behaviour

2 Boundary Flux

Boundary flux:

$$\Phi_{\beta} = I \cdot n.$$

3 Horizon Flux

Horizon flux:

$$\Phi_h = I \cdot (1 - O).$$

4 Conservation Surfaces

Conservation condition:

$$\Phi_\beta = \Phi_h.$$

5 Collapse-Transfer Modes

Collapse-transfer strength:

$$X = g(V_c, \Phi_\beta, \Phi_h).$$

6 Recurrence-Transfer Modes

Recurrence-transfer strength:

$$R = g(V_r, \Phi_\beta, \Phi_h).$$

7 Motif-Boundary Coupling

Coupling strength:

$$M = h(M_i, \Phi_\beta).$$

8 Gradient-Horizon Deformation

Deformation:

$$D = h(\nabla F, O).$$

9 Regime-Dependent Boundary–Horizon Behaviour

Behaviour mode:

$$T = q(P, \Phi_\beta, \Phi_h).$$

10 Conclusion

The Boundary–Horizon Law defines flux, conservation surfaces, collapse-transfer, recurrence-transfer, motif coupling, gradient deformation, and regime-dependent behaviour. This edition presents the basis-independent formulation suitable for academic publication and cross-domain application.